

ORGNZ-01 LAB: Introduction to Organizing Data in Grail - Hands-on Exercises

Series: ORGNZ — Organize Data: Buckets, Segments, Security | **Notebook:** 1 of 10 | **Type:** LAB | **Created:** February 2026 | **Last Updated:** 04/26/2026

Overview

This lab notebook contains 3 hands-on exercises extracted from **ORGNZ-01: Introduction to Organizing Data in Grail**. Complete the lecture notebook first, then work through these exercises to reinforce the concepts with real DQL queries against your Dynatrace environment.

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Prerequisites

Requirement	Details
Completed	ORGNZ-01: Introduction to Organizing Data in Grail (lecture notebook)
Dynatrace Environment	SaaS tenant with Grail enabled
Permissions	logs.read , metrics.read , entities.read , spans.read

Exercise 1: Buckets (Physical Organization)

ORGNZ-01: Introduction to Organizing Data in Grail

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Dynatrace Grail organizes data in **buckets**, **tables**, and **views** to ensure efficient storage, flexible access, and scalable querying. Understanding how to organize your data is fundamental to achieving optimal query performance, compliance requirements, and access control.

| Requirement | Details |
|---

1. Why Organize Data?
2. The Grail Data Model
3. Three Pillars of Data Organization
4. When to Use Each Mechanism

- 5. Permission Levels in Grail
 - 6. Exploring Your Data Organization
 - 7. Organization Decision Framework
-

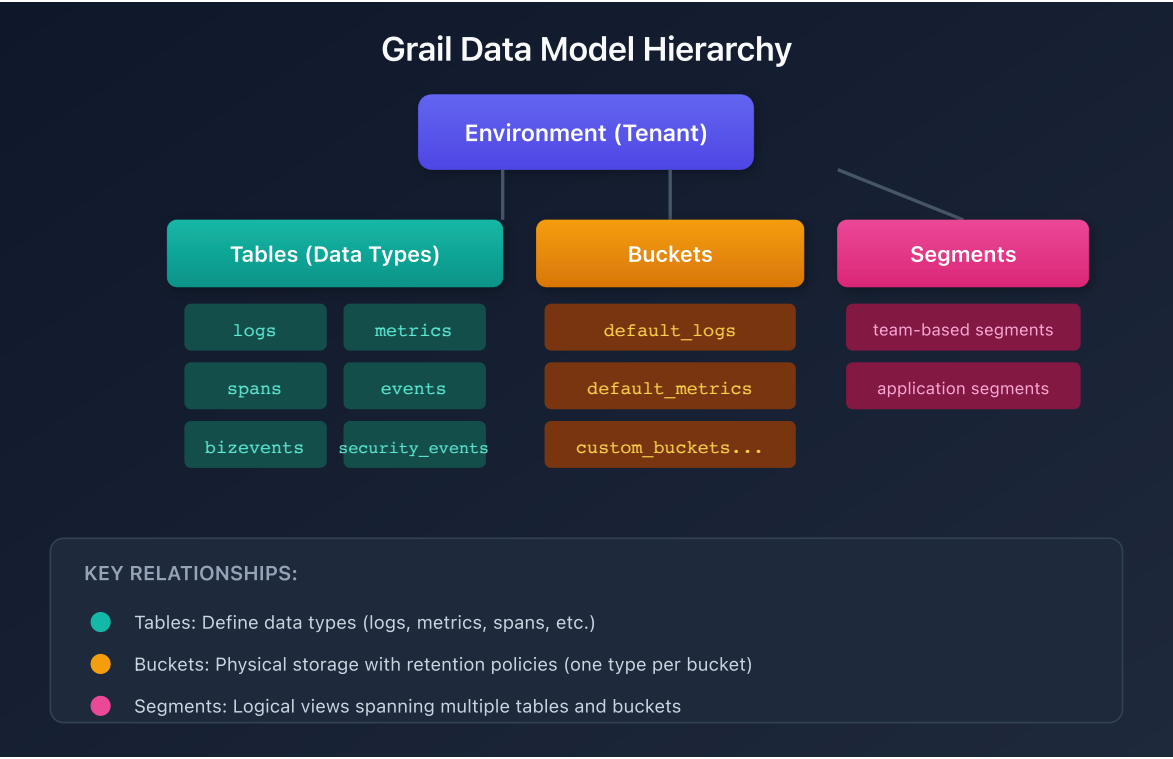
By the end of this notebook, you will:

- Understand why data organization matters in Grail
- Know the three pillars of data organization: buckets, segments, and security
- Recognize when to use each organization mechanism
- Understand the Grail data model structure

Effective data organization in Grail enables:

Benefit	Description
Query Performance	Reduce query execution time by limiting data scope
Cost Control	Transparent GiB/day pricing with bucket-level attribution
Access Control	Fine-grained permissions from bucket to field level
Compliance	Meet retention and data residency requirements
Operational Clarity	Teams see only relevant data for faster troubleshooting

Grail organizes data hierarchically with tables (data types), buckets (storage containers), and segments (logical views):



Buckets are logical storage containers where records are stored:

Aspect	Description
Purpose	Physical data partitioning and retention control
Scope	One data type per bucket (logs, metrics, etc.)
Retention	1 day to 10 years per bucket
Access	IAM policies can grant bucket-level access
Cost	Enable precise cost attribution by team/LOB

Segments provide dynamic, cross-signal data views:

Aspect	Description
Purpose	Real-time filtering without physical separation
Scope	Spans multiple data types and buckets
Flexibility	Rule-based, can use variables and relationships
Access	Governed by existing access controls
Cost	No direct storage cost impact

Security context enables fine-grained permissions:

Aspect	Description
Purpose	Record-level access control
Mechanism	<code>dt.security_context</code> attribute on records
Flexibility	Custom values, hierarchical encoding
Enforcement	IAM policies filter at query time

Need	Solution	Rationale
Different retention periods	Buckets	Retention is set per bucket
Cost attribution by team	Buckets	Direct GiB/day billing visibility
Broad team isolation	Buckets	Simple bucket-level IAM policies
Dynamic data views	Segments	Filter across data types at query time
Application-centric views	Segments	Group related entities and signals
Record-level access control	Security Context	Fine-grained ABAC
Multi-team shared buckets	Security Context	Filter records within same bucket

Grail supports permissions at multiple levels:

Level	Granularity	Example Use Case
Bucket	All records in a bucket	Team owns entire bucket
Table	All records of a data type	Access to all logs
Record	Individual records by attribute	By host group, namespace, security context
Field	Specific fields on records	Mask sensitive fields

Important: Without permissions, users cannot query data from Grail. Permissions must be explicitly granted.

```
// List all buckets in your environment
fetch dt.system.buckets
| fields name, display_name, dt.system.table, retention_days,
estimated_uncompressed_bytes
| sort dt.system.table asc, name asc
```

Exercise 2: Buckets (Physical Organization)

```
// Summarize data volume by bucket
fetch logs, from:-1h
| summarize recordCount = count(), by:{dt.system.bucket}
| sort recordCount desc
```

Exercise 3: Buckets (Physical Organization)

```
// Check for records with security context
fetch logs, from:-1h
| filter isNotNull(dt.security_context)
| summarize count = count(), by:{dt.security_context}
| sort count desc
| limit 20
```

Lab Summary

You have completed 3 hands-on exercises for **ORGNZ-01: Introduction to Organizing Data in Grail**.

Exercises Completed

- ☐ Exercise 1: Buckets (Physical Organization)
- ☐ Exercise 2: Buckets (Physical Organization)
- ☐ Exercise 3: Buckets (Physical Organization)

Next Steps

Continue with **ORGNZ-02** for the next notebook.

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